

Trivial Graph

Graph Theory Explorer — Study Notes

Definition

A trivial graph is a graph that consists of exactly one vertex and no edges. It is the smallest possible graph, denoted $G = (V, E)$ where $|V| = 1$ and $E = \emptyset$.

Key Properties

- Contains exactly one vertex and zero edges.
- The single vertex has degree 0.
- It is considered both connected (trivially, since there's nothing to disconnect) and acyclic.
- It is a special case of the null graph where $n = 1$.
- It has exactly one connected component (itself).

Example

Let $V = \{A\}$ and $E = \emptyset$. The graph consists of a single point A , sitting alone with no connections. This is the simplest graph that can be drawn.

Why It Matters

The trivial graph is frequently used as the base case in inductive proofs about graphs (for example, proving properties of trees by induction on the number of vertices). It represents the smallest unit from which larger, more complex graphs can be built by adding vertices and edges one at a time.